

Lipschitz Derivation for GCN

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1 Converting To Simplified Form

First, our goal is to get to the Lipschitz constant form:

$$\frac{\|X^{(L)} - X_p^{(L)}\|_2}{\|\Delta - \Delta_p\|_2} \leq L_2$$

where L_2 is the Lipschitz constant.

From *An Optimization Based Framework...* we are given:

$$\|X^{(L)} - X_p^{(L)}\|_F \leq \sqrt{d}L\|\Delta - \Delta_p\|_2 \prod_{l=1}^L \|\theta^{(l)}\|_2$$

Here are the following definitions:

Δ = the normalized adjacency matrix

$X^{(l)}$ = the embeddings at the l th layer

d = dimensionality of input graph

L = the number of layers in the GCN model

p indicates *perturbation*

We know for a general matrix PQ :

$$\|PQ\|_F \leq \|P\|_F \|Q\|_2$$

We can use this to our advantage by doing:

$$\|I_d(X^{(L)} - X_p^{(L)})\|_F \leq \|I_d\|_F \|X^{(L)} - X_p^{(L)}\|_2$$

We know that $\|I_d\|_F = \sqrt{d}$.

Therefore, we can rewrite this as (*Note: I'm a little shaky about this substitution because of the $\leq$*):

$$\sqrt{d}\|X^{(L)} - X_p^{(L)}\|_2 \leq \sqrt{d}L\|\Delta - \Delta_p\|_2 \prod_{l=1}^L \|\theta^{(l)}\|_2$$

We can simplify to get:

$$\frac{||X^{(L)} - X_p^{(L)}||_2}{||\Delta - \Delta_p||_2} \leq L \prod_{l=1}^L ||\theta^{(l)}||_2$$

This was the initial form that we wanted. Therefore, we know that

$$L_2 = L \prod_{l=1}^L ||\theta^{(l)}||_2$$